EC5.407 Wireless Communications - Assignment 2

Detection and Diversity in AWGN and Fading Channels

Ande Karthik 2023102009

Code: `https://github.com/KarthiK-Ande-22/Wireless-Assignment-2`

1 Simulation set-up (common to all problems)

1.1 Model and conventions

All three problems use the conventions of the "Getting Started" section of the assignment. The constellation has unit average energy, so the SNR is set only by the noise variance:

$$E_s = \mathbb{E}|x|^2 = \frac{1}{M}\sum_{i=1}^{M}|u_i|^2 = 1, \qquad \mathbb{E}|w|^2 = \frac{1}{\mathrm{SNR}}, \qquad \mathrm{SNR} = \frac{E_s}{\mathbb{E}|w|^2}, \qquad \mathrm{SNR} = 10^{\mathrm{SNR_{dB}}/10}.$$

- **Constellations:** BPSK $\{+1, -1\}$ and 4-QAM $\{\pm A \pm jA\}$ with $A = 1/\sqrt{2}$. Both have unit energy, so they are compared at equal symbol energy.
- **Noise:** $w = \frac{1}{\sqrt{2\mathrm{SNR}}}(w_I + jw_Q) \sim \mathcal{CN}(0, 1/\mathrm{SNR})$, with $w_I, w_Q \sim \mathcal{N}(0, 1)$ independent.
- **Rayleigh fading:** $h = \frac{1}{\sqrt{2}}(h_I + jh_Q) \sim \mathcal{CN}(0, 1)$, a new value for every symbol (for every Alamouti block in Problem 3), known at the receiver.
- **Monte Carlo size:** $N = 10^6$ symbols per SNR point, for every curve and every L.
- **Seed:** set once at the start of every script with `np.random.default_rng(2026)`.
- **Vectorised:** all N symbols of an SNR point are processed at once. The only loops run over SNR points, schemes and L.

1.2 Code structure

Every block of the link in Fig. 1 of the assignment is written once, in `common/`, and every problem imports it. The code is Python 3.12 with NumPy, SciPy (only `erfc`) and Matplotlib. No library function for modulation, detection or error-rate calculation is used.

File	What it does
`constellation.py`	BPSK and 4-QAM, normalised to $E_s = 1$.
`channel.py`	Complex Gaussian samples; AWGN, Rayleigh, $1 \times L$ SIMO and 2×1 MISO channels.
`detector.py`	One ML (minimum-distance) detector for any constellation, $\hat{\imath} = \arg\min_m \lvert r - g\,u_m\rvert^2$, computed for all N samples at once as an $N \times M$ distance array.
`combining.py`	Receive combiners (none, MRC, SC) that turn the received samples into one scalar decision statistic.
`simulation.py`	Algorithm 1: draw indices, send through the channel, combine, detect, and count errors by comparing indices, $n_{\mathrm{err}} = \sum_n \mathbf{1}\{\hat{\imath}_n \neq i_n\}$.
`theory.py`	$Q(a) = \frac{1}{2}\,\mathrm{erfc}(a/\sqrt{2})$ as its own function, the formulas of the assignment table, and the exact 4-QAM SER of every scheme (Sec. 1.4).
`plotting.py`	Figures (only points with $n_{\mathrm{err}} \geq 10$), CSV tables, SNR gaps, slopes and z-scores.
`analysis.py`	
`problem1/ to problem3/`	One script per task. Each writes its figures, CSV table and a `summary.txt` with every number quoted below to `results/q1` – `results/q3`.
`overall_check.py`	Compares every plotted point of all problems with its exact theory (Sec. 1.5).

One detector for every receiver. Every receiver in this assignment reduces what it receives to an equivalent scalar channel

$$r = g\,x + n, \qquad n \sim \mathcal{CN}(0, 1/\mathrm{SNR}), \qquad g \text{ known at the receiver.} \tag{1}$$

The ML detector of Problem 1 is then applied without change; only the gain g differs: $g = 1$ (AWGN), $g = h$ (fading), $g = \|\mathbf{h}\|$ (MRC and beamforming), $g = h_{\ell^\star}$ (SC) and $g = \|\mathbf{h}\|/\sqrt{2}$ (Alamouti). The MRC and Alamouti outputs are divided by $\|\mathbf{h}\|$ so that the noise variance is $1/\mathrm{SNR}$ in every case. Without this division the gains would be $\|\mathbf{h}\|^2$ and $\|\mathbf{h}\|^2/\sqrt{2}$, with noise variance $\|\mathbf{h}\|^2/\mathrm{SNR}$. The SNR and the decisions would be the same (Sec. 3.1).

Reproducing. Each script is run as a module from the code root, for example `python3 -m problem2.receive_diversity`; the README lists all commands. Each script takes under a minute and reproduces its figures and numbers exactly.

1.3 How accurate is a simulated point?

A simulated point is $\hat{P}_\mathrm{e} = n_\mathrm{err}/N$. Each symbol is an independent Bernoulli(P_e) trial, so $\mathrm{std}(\hat{P}_\mathrm{e}) = \sqrt{P_\mathrm{e}(1-P_\mathrm{e})/N}$, and the relative accuracy is about $1/\sqrt{n_\mathrm{err}}$:

errors observed, n_err	10	100	1000	10 000
relative accuracy	±32%	±10%	±3%	±1%

A difference between simulation and theory is measured in units of this standard deviation,

$$z = \frac{\hat{P}_\mathrm{e} - P_\mathrm{e}}{\sqrt{P_\mathrm{e}(1-P_\mathrm{e})/N}}. \tag{2}$$

If the simulation is correct, z behaves like a standard normal variable. Most points then have $|z| < 2$, and the worst of a few hundred points is close to 3.

1.4 Exact 4-QAM theory used for checking

The assignment table gives an exact formula for BPSK in fading but only an approximation for 4-QAM, and no formula for Problems 2 and 3. To check every curve against an exact value, I compute the exact 4-QAM SER of each scheme in three steps.

1. **Every scheme is an AWGN channel with a random gain.** By (1), the SNR after detection is $\gamma\,\mathrm{SNR}$ with $\gamma = |g|^2$ (table below).
2. **Error probability for a fixed gain.** 4-QAM is two independent BPSK decisions (I and Q), each with half the energy: $P_\mathrm{e}(\gamma) = 2Q\big(\sqrt{\gamma\mathrm{SNR}}\big) - Q^2\big(\sqrt{\gamma\mathrm{SNR}}\big)$.
3. **Average over the gain.** Craig's forms of the Q-function,

$$Q(x) = \frac{1}{\pi}\int_0^{\pi/2} e^{-x^2/(2\sin^2\theta)}\,d\theta, \qquad Q^2(x) = \frac{1}{\pi}\int_0^{\pi/4} e^{-x^2/(2\sin^2\theta)}\,d\theta,$$

turn the average into one that only needs the moment-generating function $M_\gamma(s) = \mathbb{E}[e^{-s\gamma}]$:

$$P_\mathrm{e} = \frac{2}{\pi}\int_0^{\pi/2} M_\gamma\Big(\frac{\mathrm{SNR}}{2\sin^2\theta}\Big)d\theta - \frac{1}{\pi}\int_0^{\pi/4} M_\gamma\Big(\frac{\mathrm{SNR}}{2\sin^2\theta}\Big)d\theta. \tag{3}$$

Scheme	Gain $\gamma = \lvert g\rvert^2$	Distribution	$M_\gamma(s)$
Single antenna	$\lvert h\rvert^2$	Exp(1)	$1/(1+s)$
MRC (L), beamforming ($L=2$)	$\lVert\mathbf{h}\rVert^2$	Gamma$(L,1)$	$(1+s)^{-L}$
SC (L)	$\max_\ell \lvert h_\ell\rvert^2$	same as $\sum_{k=1}^{L} E_k/k$, $E_k \sim$ Exp(1)	$\prod_{k=1}^{L} k/(k+s)$
Alamouti	$\lVert\mathbf{h}\rVert^2/2$	$\frac{1}{2}$ Gamma$(2,1)$	$(1+s/2)^{-2}$

The integrals in (3) are evaluated with 256-point Gauss–Legendre quadrature (`ser_qam4_fading_mgf` in `theory.py`). I checked them in two ways:

- For one antenna, the result equals the closed form derived the same way, $P_e = (1-\mu) - \frac{1}{4}\left[1 - \frac{4}{\pi}\,\mu \arctan(1/\mu)\right]$ with $\mu = \sqrt{(\mathrm{SNR}/2)/(1+\mathrm{SNR}/2)}$.
- For MRC and SC with $L = 2, 4$, the result agrees with direct numerical integration over the density of γ to machine precision.

1.5 Overall check of all simulations

The script `overall_check.py` compares every plotted point ($n_{\mathrm{err}} \geq 10$) of Problems 1–3 with its exact value p, using the z-score (2), written with error counts as

$$z = \frac{n_{\mathrm{err}} - Np}{\sqrt{Np(1-p)}}.$$

If the simulation is unbiased, z behaves like a standard normal variable.

Three curves use exactly the same random numbers: MRC and SC with $L = 1$ (Problem 2), and the single-antenna curve of Problem 3. Both scripts start from seed 2026 and draw the symbols, channels and noise in the same order. Their error counts are identical, so they are counted *once*. This leaves 160 independent points (Table 1).

Table 1: Agreement between every simulated curve and its exact theory (`results/overall_check.txt`). "Max $|z|$" is the largest deviation over the plotted points of the curve. * This row also stands for SC, $L = 1$ and the single-antenna curve of Problem 3 (same random numbers, identical error counts).

Problem	Curve	Points	Max $\lvert z\rvert$	Problem	Curve	Points	Max $\lvert z\rvert$
1	BPSK, AWGN	10	2.59	2	MRC, $L = 4$	8	2.01
1	4-QAM, AWGN	13	1.95	2	SC, $L = 2$	13	1.64
1	BPSK, fading	21	3.02	2	SC, $L = 4$	9	1.12
1	4-QAM, fading	21	1.78	3	Beamforming	13	2.06
2	MRC, $L = 1$ *	13	1.85	3	Alamouti	13	2.55
2	MRC, $L = 2$	13	1.03	3	MRC, $L = 2$	13	1.85

Table 2 compares the statistics of all 160 z-values with those of a standard normal variable.

Table 2: Statistics of z over all 160 independent plotted points.

Statistic over all 160 points	Measured	Expected if unbiased
points with $\lvert z\rvert < 1$	71.2%	68.3%
points with $\lvert z\rvert < 2$	95.0%	95.4%
points with $\lvert z\rvert < 3$	99.4%	99.7%
largest $\lvert z\rvert$	3.02	reached by chance in 33% of runs
mean of z (bias)	+0.079	0 ± 0.08
mean of z^2	0.97	1 ± 0.11

Conclusion. Every simulated curve agrees with its exact theory, and the 160 deviations are spread exactly like Monte Carlo noise should be. The share of points within 1σ, 2σ and 3σ matches the normal distribution, the mean of z shows no bias, and the mean of z^2 is 1 within its accuracy. The largest deviation, 3.02σ (BPSK in fading at 6 dB, +1.3%), is the only point beyond 3σ. For 160 independent points, a maximum this large happens by chance in about one run in three. The only *systematic* difference in this report is between the 4-QAM fading simulation and the *approximate* formula of the assignment table, explained in Sec. 2.3.1.

2 Problem 1 — SER in AWGN and Rayleigh fading

Problem 1 measures the SER of BPSK and 4-QAM over the channel

$$y = hx + w, \qquad x \in \mathcal{X}, \qquad w \sim \mathcal{CN}(0, 1/\mathrm{SNR}), \tag{4}$$

with $h = 1$ in Task 1 (AWGN) and $h \sim \mathcal{CN}(0, 1)$ in Task 2 (Rayleigh fading). Both tasks use the unit-energy constellations, the ML detector and the error counting of Sec. 1. In both channels the receiver knows h, so the ML decision is

$$\hat{\imath} = \arg\min_i |y - h\,u_i|^2. \tag{5}$$

Each task below first derives the theoretical SER, then describes the simulation and shows its figure. The results are discussed in Sec. 2.3.

2.1 Task 1: AWGN channel

Theoretical expressions. With $h = 1$ the receiver sees $y = x + w$. The real and imaginary parts of w are independent, each $\mathcal{N}(0, 1/(2\mathrm{SNR}))$. The Gaussian tail function $Q(a) = \frac{1}{2}\,\mathrm{erfc}(a/\sqrt{2})$ is implemented as its own function (`qfunc`).

- **BPSK.** The points ± 1 are separated by the threshold $\mathrm{Re}\{y\} = 0$, at distance 1 from each point. An error happens when the real part of the noise is larger than 1 in the wrong direction:

$$P_{\mathrm{e}}^{\mathrm{BPSK}} = \Pr\left[\mathcal{N}(0, \tfrac{1}{2\mathrm{SNR}}) > 1\right] = Q(\sqrt{2\mathrm{SNR}}). \tag{6}$$

- **4-QAM.** The points $(\pm 1 \pm j)/\sqrt{2}$ are two BPSK signals, one on I and one on Q, each with amplitude $1/\sqrt{2}$ (half the energy) and its own independent noise. One component is wrong with probability $q = Q((1/\sqrt{2})/\sqrt{1/(2\mathrm{SNR})}) = Q(\sqrt{\mathrm{SNR}})$. The symbol is correct only if both components are correct:

$$P_{\mathrm{e}}^{\mathrm{4QAM}} = 1 - (1-q)^2 = 2Q(\sqrt{\mathrm{SNR}}) - Q^2(\sqrt{\mathrm{SNR}}). \tag{7}$$

  An error in one or in both components counts as one symbol error.

Simulation. The script `problem1/awgn_ser.py` runs Algorithm 1 for $\mathrm{SNR}_{\mathrm{dB}} \in \{0, 1, \ldots, 12\}$. For each modulation and SNR it draws $N = 10^6$ equiprobable indices, sends $x = u_i$ through $y = x + w$, detects with (5) and counts the errors. The expressions (6) and (7) are evaluated at the same SNR points. Fig. 1 shows both schemes on one figure: simulation as markers, theory as lines.

2.2 Task 2: Rayleigh fading channel

Channel. A new $h \sim \mathcal{CN}(0, 1)$ is drawn for every symbol and is known at the receiver (coherent detection). For a given h, the link is an AWGN link whose SNR is multiplied by the channel power gain $g = |h|^2$. Since $h \sim \mathcal{CN}(0, 1)$, g is exponential, $f_g(g) = e^{-g}$ for $g \geq 0$, with $\mathbb{E}[g] = 1$. So SNR is also the average received SNR.

Theoretical expressions. Each fading result is the AWGN result for a fixed g, averaged over the distribution of g.

- **BPSK.** For a fixed g, (6) becomes $Q(\sqrt{2g\mathrm{SNR}})$. Averaging over g gives:

$$P_{\mathrm{e}}^{\mathrm{BPSK}} = \int_0^\infty Q(\sqrt{2g\mathrm{SNR}})\, e^{-g}\, dg = \frac{1}{2}\Big(1 - \sqrt{\frac{\mathrm{SNR}}{1+\mathrm{SNR}}}\Big). \tag{8}$$

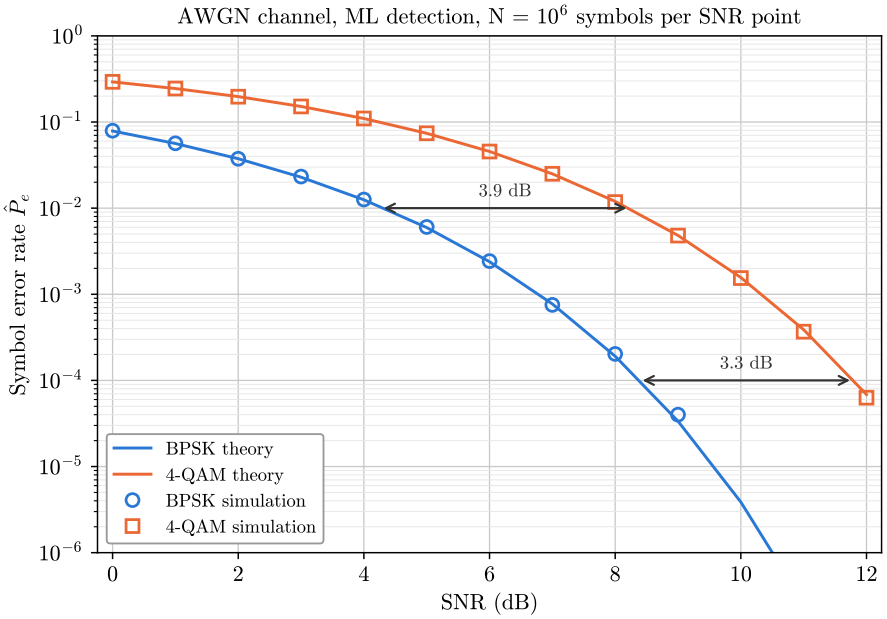


Figure 1: Problem 1, Task 1: SER of BPSK and 4-QAM in the AWGN channel ($h = 1$) with ML detection. Markers: Monte Carlo simulation with $N = 10^6$ symbols per SNR point, $\mathrm{SNR_{dB}} \in \{0, 1, \ldots, 12\}$; only points with $n_{\mathrm{err}} \geq 10$ are shown. Lines: theory (6) and (7). Arrows: the measured SNR gap between the two schemes, 3.9 dB at 10^{-2} and 3.3 dB at 10^{-4}. **The BPSK–4-QAM gap is not constant: it shrinks towards 3 dB as the SNR grows** (Sec. 2.3.3).

- **4-QAM, formula of the assignment table.** Twice the BPSK result at half the SNR:

$$P_{\mathrm{e}}^{4\mathrm{QAM}} \approx 2 \cdot \frac{1}{2}\Big(1 - \sqrt{\frac{\mathrm{SNR}/2}{1 + \mathrm{SNR}/2}}\Big). \tag{9}$$

  This is the probability of an I error plus the probability of a Q error. It leaves out the event that both are wrong, so it is an approximation (an upper bound).
- **4-QAM, exact (extra, for checking).** Averaging the complete AWGN expression (7), with $q = Q(\sqrt{g\mathrm{SNR}})$, over g gives, with Craig's formula (Sec. 1.4),

$$P_{\mathrm{e}}^{4\mathrm{QAM}} = \int_0^\infty \left[2q - q^2\right] e^{-g}\, dg = (1 - \mu) - \tfrac{1}{4}\Big[1 - \tfrac{4}{\pi}\,\mu \arctan\tfrac{1}{\mu}\Big], \qquad \mu = \sqrt{\frac{\mathrm{SNR}/2}{1 + \mathrm{SNR}/2}}. \tag{10}$$

  The first term, $1 - \mu$, is exactly the table formula (9); the second is the dropped "both wrong" term. This curve is only used to explain the simulated 4-QAM points; the comparison asked for in Task 2 uses (9).

Simulation. The script `problem1/rayleigh_ser.py` runs Algorithm 1 for $\mathrm{SNR_{dB}} \in \{0, 2, \ldots, 40\}$, with a fresh h and w for every symbol and $N = 10^6$ symbols per modulation and SNR point. The detector (5) uses the known h. Fig. 2 shows the simulation with the expressions (8)–(10). Fig. 3 is an extra figure that places the AWGN theory of Task 1 next to the fading results; it is used in the discussion of Sec. 2.3.2.

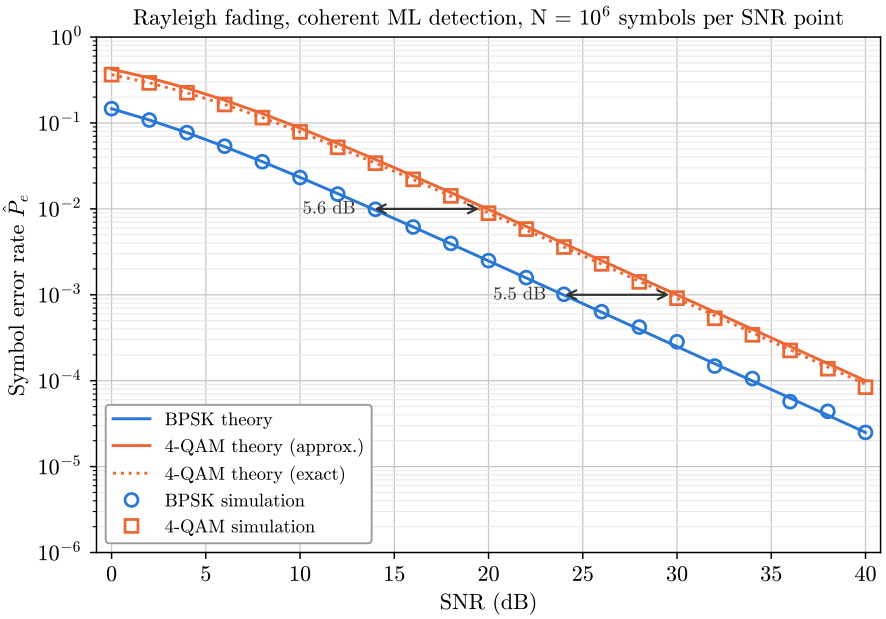


Figure 2: Problem 1, Task 2: SER of BPSK and 4-QAM in Rayleigh fading with coherent ML detection. Markers: simulation with $N = 10^6$ symbols per SNR point, $\mathrm{SNR}_{\mathrm{dB}} \in \{0, 2, \ldots, 40\}$, points with $n_{\mathrm{err}} \geq 10$. Solid lines: assignment table (8) and (9) (an approximation for 4-QAM). Dotted: exact 4-QAM SER (10). Arrows: the measured BPSK–4-QAM gap, 5.6 dB at 10^{-2} and 5.5 dB at 10^{-3}. **The gap is nearly constant, about 5.5 dB: both curves are parallel lines of slope −1** (Sec. 2.3.3).

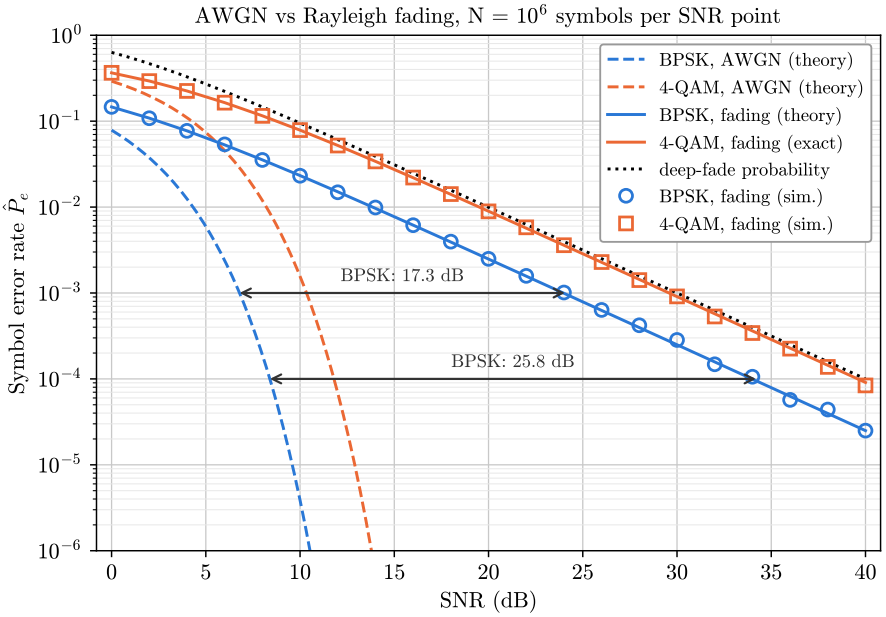


Figure 3: Extra figure: AWGN (dashed, theory) against Rayleigh fading (solid lines: theory; markers: the simulation of Fig. 2, $N = 10^6$ symbols per SNR point, 0–40 dB). Dotted: deep-fade probability $\Pr[|h|^2\mathrm{SNR} < 1] = 1 - e^{-1/\mathrm{SNR}} \approx 1/\mathrm{SNR}$. Arrows: the extra SNR that BPSK needs in fading, 17.3 dB at 10^{-3} and 25.8 dB at 10^{-4}. **The AWGN–fading gap is not constant: it grows as the target SER falls** (Sec. 2.3.3).

2.3 Discussion

The assignment asks for two points: the agreement between simulation and theory, with the reason for any deviation (Sec. 2.3.1), and the difference between the AWGN and fading curves at high SNR, explained with the deep-fade argument (Sec. 2.3.2). Sec. 2.3.3 explains the gaps marked by the arrows in Figs. 1–3.

2.3.1 Agreement between simulation and theory

- **AWGN (Fig. 1, Table 7).** Both schemes match theory: max $|z|$ is 2.59 (BPSK) and 1.95 (4-QAM). The largest *relative* deviations occur where few errors are seen. For example, BPSK at 9 dB has 40 errors where 33.6 are expected (+19%, only 1.1σ). At 10, 11 and 12 dB, BPSK gives 6, 0 and 0 errors (3.9, 0.26 and 0.01 expected). These points are not plotted (G.7): 10^6 symbols cannot measure such small error rates.
- **Fading, BPSK.** The simulation follows the exact formula $\frac{1}{2}(1 - \sqrt{\mathrm{SNR}/(1+\mathrm{SNR})})$, within $\pm 2.5\%$ wherever $n_{\mathrm{err}} > 1000$. With fewer errors the scatter grows, as the table of Sec. 1.3 predicts. The largest relative deviation is +14% at 30 dB (284 errors against 250 expected, 2.2σ).
- **Fading, 4-QAM: a systematic deviation from the table formula.** The simulated SER is *below* the table formula at every SNR: 13.5% lower at 0 dB and 7–11% lower from 10 to 30 dB. That is up to 116σ, so it is not noise. The simulation is not wrong: it matches the *exact* curve (dotted in Fig. 2, max $|z| = 1.78$). The reasons:
  - Per symbol, the error probability is $2q - q^2$ with $q = Q(\sqrt{|h|^2\mathrm{SNR}})$ the error probability of one component. The table formula keeps only $\mathbb{E}[2q]$ and drops $-\mathbb{E}[q^2]$, the event that I *and* Q are both wrong. It is therefore an upper bound.
  - In AWGN, $q^2 \ll q$ at high SNR, so the dropped term does not matter. In fading it does: errors happen mostly in deep fades, and there both components see the same small $|h|$ and fail together.
  - At high SNR, $\mathbb{E}[2q] \to 1/\mathrm{SNR}$ and $\mathbb{E}[q^2] \to \frac{1}{4}(1 - \frac{2}{\pi})/\mathrm{SNR} = 0.091/\mathrm{SNR}$. The exact SER is therefore $0.909/\mathrm{SNR}$, and the table formula is 10% too high. The simulation output confirms this: the formula exceeds the exact value by 15.8% at 0 dB and by 10.0% from 30 dB on.
  - On the dB axis this is only $10\log_{10}(1/0.909) = 0.41$ dB, which is why the two lines look so close in Fig. 2.

2.3.2 AWGN versus fading at high SNR: the deep-fade argument

What the figures show.

- *AWGN:* the SER falls *exponentially* with SNR, because $Q(x) \le \frac{1}{2}e^{-x^2/2}$. The curves are a steep "waterfall": BPSK goes from 10^{-3} at 6.8 dB to 10^{-4} at 8.4 dB, one decade in only 1.6 dB.
- *Rayleigh fading:* the curves become straight lines of slope -1, so $P_{\mathrm{e}} \propto \mathrm{SNR}^{-1}$: one decade per 10 dB. The fitted slope over 30–40 dB is 1.02 for both schemes. At high SNR, $P_{\mathrm{e}}^{\mathrm{BPSK}} \approx 1/(4\mathrm{SNR})$ and $P_{\mathrm{e}}^{4\mathrm{QAM}} \approx 0.909/\mathrm{SNR}$.

Why: errors come from deep fades.

1. For a given channel, the link is an AWGN link whose SNR is scaled by $|h|^2$: $P_{\mathrm{e}}(h) = Q(\sqrt{2|h|^2\mathrm{SNR}})$ for BPSK.
2. This gives two regimes:
   - if $|h|^2\mathrm{SNR} \gg 1$, $P_{\mathrm{e}}(h)$ is exponentially small, so good channels contribute almost no

errors;

- if $|h|^2\text{SNR} \lesssim 1$ (a *deep fade*), $P_\text{e}(h)$ is of order one (at least $Q(\sqrt{2}) = 0.079$).

3. Because $|h|^2 \sim \text{Exp}(1)$, a deep fade is not rare:

$$\Pr\left[|h|^2 < 1/\text{SNR}\right] = 1 - e^{-1/\text{SNR}} \approx \frac{1}{\text{SNR}}.$$

   This is the dotted line in Fig. 3. It runs right along the fading curves, with the same slope.

4. Hence $P_\text{e} \approx \Pr[\text{deep fade}] \times O(1) \propto 1/\text{SNR}$. More transmit power makes deep fades only proportionally rarer, so the SER also falls only linearly.

Simulation evidence. The Problem 1 script also records $|h|^2\text{SNR}$ for every BPSK symbol and checks which errors happen during a deep fade (`results/q1/rayleigh_summary.txt`):

SNR_dB	10	20	30	40
symbols sent in a deep fade	9.5%	1.0%	0.10%	0.010%
errors (out of 10^6 symbols)	23 570	2 506	236	24
errors that happened in a deep fade	77%	74%	70%	83%

Only about 1/SNR of the symbols are sent in a deep fade, yet these few symbols cause about three quarters of all errors at every SNR.

The price of fading. SNR needed for a target SER:

Scheme	Target P_e	AWGN (theory)	Rayleigh (simulation)	Extra SNR needed
BPSK	10^{-3}	6.8 dB	24.1 dB	17.3 dB
BPSK	10^{-4}	8.4 dB	34.2 dB	25.8 dB
4-QAM	10^{-3}	10.4 dB	29.6 dB	19.2 dB
4-QAM	10^{-4}	11.8 dB	39.3 dB	27.5 dB

The penalty grows as the target SER falls (arrows in Fig. 3). Reducing it is the purpose of the diversity schemes of Problems 2 and 3.

2.3.3 Gaps between the curves (arrows in the figures)

- **BPSK versus 4-QAM in AWGN: not constant** (Fig. 1). Measured: 3.87 dB at 10^{-2}, 3.54 dB at 10^{-3} and 3.30 dB at 10^{-4} (theory 3.89, 3.56 and 3.40 dB).
  - At equal symbol energy, each 4-QAM component has half the energy of a BPSK symbol: $Q(\sqrt{\text{SNR}})$ against $Q(\sqrt{2\text{SNR}})$. That alone is a shift of exactly 3 dB.
  - 4-QAM also has the factor 2 in front of $Q(\cdot)$ (either component can fail). On a steep waterfall this factor costs only a small extra SNR, which shrinks as the SNR grows, so the gap tends to 3.01 dB.
- **BPSK versus 4-QAM in fading: constant, about 5.6 dB** (Fig. 2). Measured: 5.56 dB at 10^{-2} and 5.53 dB at 10^{-3} (exact theory 5.66 and 5.61 dB). At 10^{-4} the measured value, 5.11 dB, is less accurate: there the 4-QAM curve is read between its last two points, with only 138 and 84 errors.
  - Both curves have slope -1, so at high SNR they are parallel, $P_\text{e} \approx c/\text{SNR}$, and only the constant c depends on the modulation: $c = 0.25$ for BPSK and 0.909 for 4-QAM. The offset is $10\log_{10}(0.909/0.25) = 5.61$ dB.
  - The offset is larger than in AWGN because on a slope-1 line a factor in front of the SER is never small. Half the energy per component doubles c (3.01 dB), two components double it again (3.01 dB), and the joint-error term removes 9% (-0.41 dB).
- **AWGN versus fading: not constant, it grows** (Fig. 3). For BPSK it is 17.3 dB at 10^{-3} and 25.8 dB at 10^{-4}. One curve falls exponentially and the other as 1/SNR, so they cannot be parallel.

3 Problem 2 — Receive diversity (SIMO $1 \times L$), 4-QAM

Problem 2 adds L receive antennas to the fading channel of Problem 1. One transmit antenna sends a 4-QAM symbol x, and the receiver collects

$$\mathbf{y} = \mathbf{h}x + \mathbf{w}, \qquad \mathbf{h} = (h_1, \dots, h_L)^\top, \qquad \mathbf{w} = (w_1, \dots, w_L)^\top. \tag{11}$$

- **Channel:** $h_\ell \sim \mathcal{CN}(0,1)$, independent across antennas and drawn again for every symbol. The receiver knows $\mathbf{h}$.
- **Noise:** $w_\ell \sim \mathcal{CN}(0, 1/\mathrm{SNR})$, independent across antennas and independent of $\mathbf{h}$.
- **SNR:** since $\mathbb{E}|h_\ell|^2 = \mathbb{E}|x|^2 = 1$, SNR is the average SNR *per receive antenna*. For $L = 1$ the model is exactly the fading channel of Problem 1.
- **Code:** `simo_rayleigh_channel` (`common/channel.py`) draws $\mathbf{h}$ and $\mathbf{w}$ as $N \times L$ arrays; the constellation and the ML detector of Problem 1 are reused unchanged.

Each receiver turns the L samples into one scalar decision statistic $r = g\,x + n$ (1). Task 1 derives r for MRC and SC, Task 2 simulates both, and the results are discussed in Sec. 3.3.

3.1 Task 1: combining schemes

Both receivers reduce $\mathbf{y}$ to an equivalent scalar channel and pass (r, g) to the Problem 1 detector $\hat{\imath} = \arg\min_i |r - g\,u_i|^2$. The table summarises the result; the derivations follow.

Receiver	Decision statistic r	Gain g	Noise variance	Output SNR
MRC (as implemented)	$\mathbf{h}^H\mathbf{y}/\lVert\mathbf{h}\rVert$	$\lVert\mathbf{h}\rVert$	$1/\mathrm{SNR}$	$\lVert\mathbf{h}\rVert^2\mathrm{SNR}$
MRC (not normalised)	$\mathbf{h}^H\mathbf{y}$	$\lVert\mathbf{h}\rVert^2$	$\lVert\mathbf{h}\rVert^2/\mathrm{SNR}$	$\lVert\mathbf{h}\rVert^2\mathrm{SNR}$
SC	$y_{\ell^\star}$, $\ell^\star = \arg\max_\ell \lvert h_\ell\rvert^2$	$h_{\ell^\star}$	$1/\mathrm{SNR}$	$\max_\ell \lvert h_\ell\rvert^2\mathrm{SNR}$

In the code these are `mrc_combine` and `selection_combine` (`common/combining.py`).

3.1.1 Maximal ratio combining (MRC)

Combining. Each branch is multiplied by its conjugate channel gain h_ℓ^* and the branches are added:

$$\mathbf{h}^H\mathbf{y} = \sum_{\ell=1}^{L} h_\ell^*\, y_\ell = \lVert\mathbf{h}\rVert^2 x + \mathbf{h}^H\mathbf{w}.$$

The factor h_ℓ^* does two things: it removes the phase of each branch, so all signal parts add in phase, and it gives a strong branch more weight than a weak one.

Equivalent scalar channel. The code divides the sum by $\lVert\mathbf{h}\rVert$:

$$r = \frac{\mathbf{h}^H\mathbf{y}}{\lVert\mathbf{h}\rVert} = \lVert\mathbf{h}\rVert\, x + \underbrace{\frac{\mathbf{h}^H\mathbf{w}}{\lVert\mathbf{h}\rVert}}_{n}, \qquad \mathrm{Var}(n) = \frac{\sum_\ell |h_\ell|^2 \cdot (1/\mathrm{SNR})}{\lVert\mathbf{h}\rVert^2} = \frac{1}{\mathrm{SNR}}. \tag{12}$$

This is (1) with the real, positive gain $g = \lVert\mathbf{h}\rVert$ and noise variance $1/\mathrm{SNR}$. The output SNR is $\lVert\mathbf{h}\rVert^2\mathrm{SNR} = \sum_\ell |h_\ell|^2\mathrm{SNR}$: MRC adds the SNRs of all branches. The Problem 1 detector is applied as $\hat{\imath} = \arg\min_i |r - \lVert\mathbf{h}\rVert u_i|^2$.

Why this is exactly the ML rule. With L antennas the ML rule is $\arg\min_i \lVert\mathbf{y} - \mathbf{h}u_i\rVert^2$. Expanding and using $\mathbf{h}^H\mathbf{y} = \lVert\mathbf{h}\rVert\, r$:

$$\lVert\mathbf{y} - \mathbf{h}u_i\rVert^2 = \lVert\mathbf{y}\rVert^2 - 2\lVert\mathbf{h}\rVert\,\mathrm{Re}(u_i^* r) + \lVert\mathbf{h}\rVert^2|u_i|^2 = |r - \lVert\mathbf{h}\rVert u_i|^2 + \underbrace{\left(\lVert\mathbf{y}\rVert^2 - |r|^2\right)}_{\text{same for every } i}.$$

The last term does not depend on i, so both rules pick the same symbol. Without the normalisation, the statistic is $\mathbf{h}^H\mathbf{y} = \|\mathbf{h}\|^2 x + \mathbf{h}^H\mathbf{w}$, with gain $\|\mathbf{h}\|^2$ and noise variance $\|\mathbf{h}\|^2/\mathrm{SNR}$. Its metric is $|\mathbf{h}^H\mathbf{y} - \|\mathbf{h}\|^2 u_i|^2 = \|\mathbf{h}\|^2\,|r - \|\mathbf{h}\|u_i|^2$, the same metric multiplied by a positive constant, so it is also ML.

3.1.2 Selection combining (SC)

Selection. The receiver keeps only the antenna with the largest instantaneous channel gain,

$$\ell^\star = \arg\max_\ell |h_\ell|^2, \qquad r = y_{\ell^\star} = h_{\ell^\star} x + w_{\ell^\star}, \tag{13}$$

and ignores the other $L-1$ samples. In hardware this needs only one receive chain, switched to the best antenna.

Equivalent scalar channel. This is (1) with the complex gain $g = h_{\ell^\star}$, whose phase is removed by the coherent detector. The choice of $\ell^\star$ depends only on $\mathbf{h}$, which is independent of $\mathbf{w}$, so $w_{\ell^\star}$ is still $\mathcal{CN}(0, 1/\mathrm{SNR})$. The output SNR is $\max_\ell |h_\ell|^2 \mathrm{SNR}$, and the Problem 1 detector is applied to the selected branch: $\hat{\imath} = \arg\min_i |y_{\ell^\star} - h_{\ell^\star} u_i|^2$.

MRC against SC. For every channel, $\sum_\ell |h_\ell|^2 \geq \max_\ell |h_\ell|^2$, so the output SNR of MRC is never lower than that of SC. SC trades this SNR for a simpler receiver.

Numerical check (`results/q2/summary.txt`). At 10 dB with 10^6 symbols, the measured noise variance of the decision statistic, divided by the predicted $1/\mathrm{SNR}$, is 0.9986 (MRC) and 0.9987 (SC) for $L = 2$, and 0.9993 (MRC) and 1.0003 (SC) for $L = 4$. Each is within 0.15% of 1, the Monte Carlo accuracy of 10^6 samples. The mean gains $\mathbb{E}\|\mathbf{h}\|^2 = L$ and $\mathbb{E}\max_\ell |h_\ell|^2 = \sum_{k=1}^{L} 1/k$ are also reproduced to 0.01 dB.

3.2 Task 2: simulation

The script `problem2/receive_diversity.py` runs Algorithm 1 for $L \in \{1, 2, 4\}$ and $\mathrm{SNR}_{\mathrm{dB}} \in \{0, 2, \ldots, 24\}$. For each L and SNR:

1. draw $N = 10^6$ equiprobable indices and set $x = u_i$ (4-QAM);
2. draw $\mathbf{h}$ and $\mathbf{w}$ ($N \times L$ each) and form $\mathbf{y} = \mathbf{h}x + \mathbf{w}$;
3. apply MRC (12) and SC (13) to the *same* $\mathbf{y}$ and $\mathbf{h}$;
4. detect each statistic with the Problem 1 detector and count the errors of each receiver.

Because MRC and SC see the same symbols, channels and noise, any difference between their curves comes from the combining alone. Fig. 4 shows all curves on one figure, as the assignment asks: one colour per L, solid lines for MRC and dashed lines for SC. The grey band is the exact single-antenna curve of Problem 1, for the $L = 1$ consistency check, and the arrows mark the MRC–SC gap at 10^{-3}. Fig. 5 is an extra figure that compares the same points with the exact theory of Sec. 1.4.

What the figures should show. Three predictions from theory are checked in the discussion:

- $L = 1$*:* MRC and SC both reduce to the Problem 1 detector, so their curves coincide with each other and with the 4-QAM fading curve of Problem 1.
- *Diversity order:* both receivers have $P_\mathrm{e} \propto \mathrm{SNR}^{-L}$ at high SNR, so the curves should fall by about L decades per 10 dB: 1, 2 and 4.
- *Array gain:* MRC raises the mean output SNR by a factor L (3.01 and 6.02 dB), SC only by $\sum_{k=1}^{L} 1/k$ (1.76 and 3.19 dB). So for $L = 2$ and $L = 4$, MRC should lead SC by roughly 1.25 and 2.83 dB.

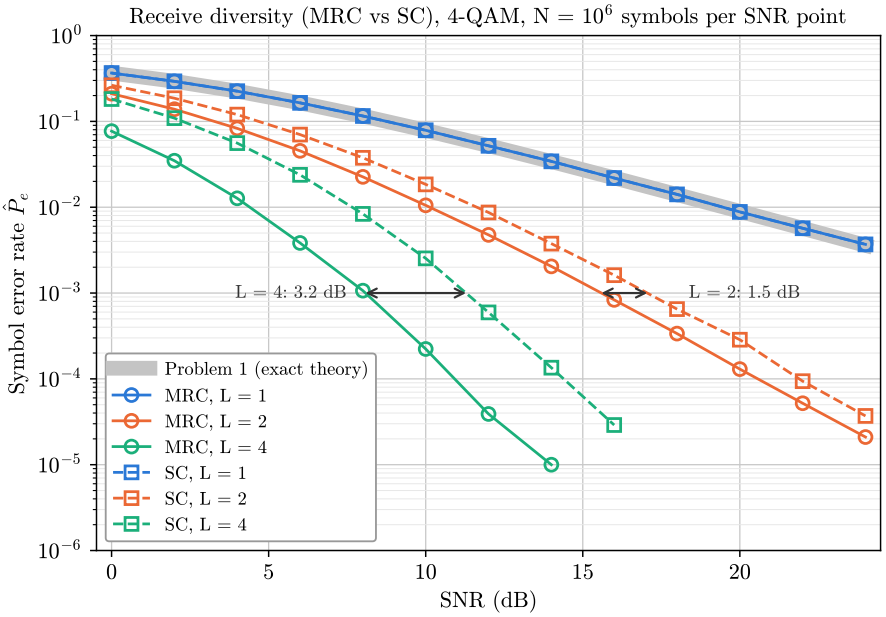


Figure 4: Problem 2: SER of 4-QAM with $L \in \{1, 2, 4\}$ receive antennas. MRC: solid lines, circles; SC: dashed lines, squares; one colour per L. Grey band: exact single-antenna 4-QAM SER of Problem 1. Monte Carlo with $N = 10^6$ symbols per SNR point, $\mathrm{SNR}_{\mathrm{dB}} \in \{0, 2, \ldots, 24\}$ (average SNR per antenna); points with $n_{\mathrm{err}} < 10$ are omitted, so the $L = 4$ curves stop at 14 and 16 dB. Arrows: the MRC–SC gap at 10^{-3}, 3.2 dB for $L = 4$ and 1.5 dB for $L = 2$. **For each L, MRC and SC have the same slope, and MRC is better by a nearly constant gap; for $L = 1$ the two curves are identical** (Sec. 3.3.4).

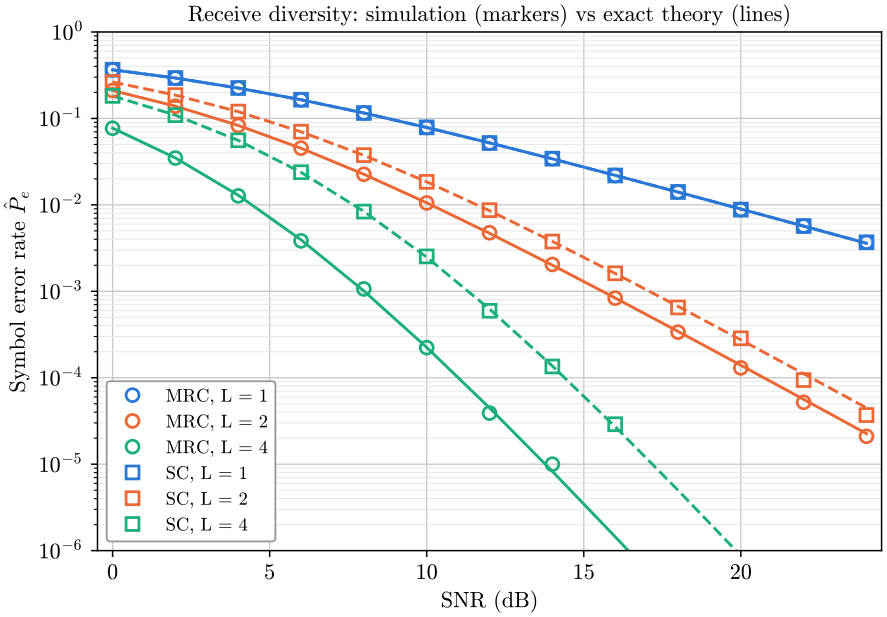


Figure 5: Extra figure: the simulated points of Fig. 4 ($N = 10^6$ symbols per SNR point, 0–24 dB) against the exact theory of Sec. 1.4 (lines; MRC solid, SC dashed). **Every plotted point lies on its theory curve** (within 2.01σ).

3.3 Discussion

The assignment asks for four points: the decision statistics of Task 1 (Sec. 3.3.1), a consistency check for $L = 1$ (Sec. 3.3.2), the high-SNR slope of each curve against the predicted diversity order (Sec. 3.3.3), and the SNR gap between MRC and SC with its origin in array gain (Sec. 3.3.4).

3.3.1 Decision statistics

These are derived in Sec. 3.1. Both receivers give $r = g\,x + n$ with $n \sim \mathcal{CN}(0, 1/\mathrm{SNR})$, so the Problem 1 detector is reused unchanged.

3.3.2 Consistency check ($L = 1$)

- With one antenna, MRC is only a phase rotation, $r = (h^*/|h|)\,y$, and SC has only one branch to choose. Both reduce to coherent detection on $y = hx + w$. They also process the same realisations, so their error counts are *identical* at every SNR (checked in the code), and the two curves coincide in Fig. 4.
- The $L = 1$ curve lies on the 4-QAM fading curve of Problem 1. It agrees with the exact theory within 1.85σ, and with an independent re-run of the Problem 1 simulation within 1.69σ.
- All MRC and SC curves agree with exact theory (Fig. 5, Table 9); every plotted point is within 2.01σ.

3.3.3 High-SNR slope versus diversity order

MRC and SC both have diversity order L, i.e. $P_\mathrm{e} \propto \mathrm{SNR}^{-L}$. An error now needs *all* L independent branches to fade at once. With $\epsilon = 1/\mathrm{SNR}$:

$$\Pr\left[\|\mathbf{h}\|^2 < \epsilon\right] \approx \frac{\epsilon^L}{L!} \quad \text{(MRC)}, \qquad \Pr\left[\max_\ell |h_\ell|^2 < \epsilon\right] = \left(1 - e^{-\epsilon}\right)^L \approx \epsilon^L \quad \text{(SC)}. \tag{14}$$

Table 3 gives the slope $-\Delta \log_{10} \hat{P}_\mathrm{e} / \Delta \log_{10} \mathrm{SNR}$ at the high-SNR end of each curve.

Table 3: High-SNR slopes over the last four points with at least 50 errors: local slope of each 2 dB step, least-squares fit through the four points, the exact theory fitted over the same points, and the exact theory at 60–70 dB.

			Fitted slope		Exact theory,	Diversity
Receiver	Fit range	Local slopes	simulation	exact theory	60–70 dB	order
MRC / SC, $L = 1$	18–24 dB	1.02, 0.95, 0.94	0.97	0.99	1.00	1
MRC, $L = 2$	16–22 dB	1.97, 2.07, 1.99	2.01	1.96	2.00	2
SC, $L = 2$	16–22 dB	1.98, 1.78, 2.42	2.03	1.93	2.00	2
MRC, $L = 4$	4–10 dB	2.60, 2.78, 3.40	2.91	2.94	4.00	4
SC, $L = 4$	8–14 dB	2.59, 3.15, 3.22	3.00	2.98	4.00	4

- **$L = 1$ and $L = 2$:** the fitted slopes (0.97, 2.01 and 2.03) are the predicted diversity orders 1 and 2. A single 2 dB step is less accurate (about ± 0.2–0.3 with a few hundred errors per point), which explains the 1.78 and 2.42 in the SC row.
- **$L = 4$:** the measured slope is about 3, not 4. This is not a contradiction: L is the *asymptotic* slope. With 10^6 symbols the $L = 4$ curves can only be measured down to about 10^{-5} (up to 10–14 dB), where they are still bending. Exact theory over the same points gives the same slopes (2.94 and 2.98). It reaches 4.00 only at 60–70 dB, where P_e is between 10^{-22} and 10^{-28}, far below anything a simulation can reach. The slope does grow from 2.6 to 3.4 along the measured part, as expected.

3.3.4 SNR gap between MRC and SC, and array gain

The arrows in Fig. 4 mark this gap. Table 4 gives the SNR each receiver needs for a target SER.

Table 4: SNR (dB) needed for a target SER, read off the simulated curves (exact theory in brackets), and the gap SC − MRC.

		$P_e = 10^{-2}$	$P_e = 10^{-3}$	$P_e = 10^{-4}$
$L = 2$	MRC	10.13 (10.14)	15.59 (15.60)	20.57 (20.73)
	SC	11.63 (11.61)	17.05 (17.09)	21.89 (22.24)
	gap	**1.50** (1.47)	**1.46** (1.49)	**1.32** (1.50)
$L = 4$	MRC	4.40 (4.46)	8.08 (8.01)	10.92 (11.02)
	SC	7.66 (7.69)	11.28 (11.34)	14.39 (14.40)
	gap	**3.26** (3.23)	**3.20** (3.33)	**3.47** (3.38)

- **The gap is almost constant for each L**: about 1.5 dB for $L = 2$ and 3.2–3.5 dB for $L = 4$. MRC and SC have the same slope L, so their curves are parallel at high SNR. (The 1.32 dB at 10^{-4} for $L = 2$ is read between points with only 52–286 errors, so it is accurate to only about ± 0.2 dB.)
- **Origin: array gain**, the gain in *average* output SNR over one antenna, $\mathbb{E}[\mathrm{SNR}_{\mathrm{out}}]/\mathrm{SNR}$:
  - MRC adds the signals of all L antennas coherently, while the noises add incoherently: $\mathbb{E}\|\mathbf{h}\|^2 = L$, i.e. 3.01 dB ($L = 2$) and 6.02 dB ($L = 4$).
  - SC throws away the energy received on the other $L-1$ antennas: $\mathbb{E}\max_\ell |h_\ell|^2 = \sum_{k=1}^{L} 1/k$, i.e. only 1.76 dB ($L = 2$) and 3.19 dB ($L = 4$).
  - The difference, 1.25 dB ($L = 2$) and 2.83 dB ($L = 4$), is most of the gap. It also explains why the gap grows with L: MRC's array gain grows like L, SC's only like $\ln L$.
- **Why the gap is a little larger than 1.25 / 2.83 dB.** At high SNR the SER depends on how often the output SNR is *small*, not on its mean. By (14), SC is in a deep fade $L!$ times as often as MRC, so its SER is $L!$ times higher. On curves of slope L that is a horizontal shift of $10\log_{10}(L!)/L$: 1.51 dB ($L = 2$) and 3.45 dB ($L = 4$). The measured gaps lie between the array-gain value and this limit.
- **Between different L, the gap is not constant.** The slopes differ (1, 2 and 4), so the curves for different L spread apart as the SER falls. This growing gap is the diversity gain.

4 Problem 3 — Transmit diversity (MISO 2×1), 4-QAM

Problem 3 moves the two antennas to the transmitter. If antenna ℓ sends s_ℓ in a channel use, the single receive antenna gets

$$y = h_1 s_1 + h_2 s_2 + w. \tag{15}$$

- **Channel:** $h_1, h_2 \sim \mathcal{CN}(0, 1)$ i.i.d., known at the receiver, and drawn again for every symbol (for every block of two channel uses in Alamouti).
- **Noise:** $w \sim \mathcal{CN}(0, 1/\mathrm{SNR})$.
- **Fair comparison:** every scheme sends a total energy $\mathbb{E}[|s_1|^2 + |s_2|^2] = 1$ per channel use, the same as one antenna sending x. So SNR is the total transmit energy divided by the noise variance.
- **Code:** `miso_channel` (`common/channel.py`) implements (15); the transmitters and receivers are in `problem3/transmit_diversity.py`. The constellation and the detector of Problem 1 are reused.

As in Problem 2, every receiver ends with one scalar channel $r = g\,x + n$, $n \sim \mathcal{CN}(0, 1/\mathrm{SNR})$ (1). Task 1 derives it for each scheme, Task 2 simulates them, and Sec. 4.3 discusses the results.

4.1 Task 1: transmission schemes

The table summarises the three schemes, with MRC ($L = 2$) of Problem 2 for comparison; the derivations follow.

Scheme	Channel known at transmitter	Symbol scaling	Gain g	Output SNR
Single antenna	no	$s = x$	h	$\lvert h\rvert^2$SNR
Beamforming	yes	$s_\ell = (h_\ell^*/\lVert\mathbf{h}\rVert)\, x$	$\lVert\mathbf{h}\rVert$	$\lVert\mathbf{h}\rVert^2$SNR
Alamouti	no	$\frac{1}{\sqrt{2}}$ on every entry of $\mathbf{S}$	$\lVert\mathbf{h}\rVert/\sqrt{2}$	$\lVert\mathbf{h}\rVert^2$SNR/2
MRC, $L = 2$	–	$s = x$ (one transmit antenna)	$\lVert\mathbf{h}\rVert$	$\lVert\mathbf{h}\rVert^2$SNR

4.1.1 Single antenna (reference)

$s = x$ is sent from one antenna, so $y = hx + w$: the fading channel of Problem 1, with $g = h$ and output SNR $|h|^2$SNR.

4.1.2 Transmit beamforming

Weights. The transmitter knows (h_1, h_2) and sends the same symbol from both antennas, antenna ℓ sending $v_\ell x$ with

$$\mathbf{v} = \frac{\mathbf{h}^*}{\lVert\mathbf{h}\rVert} = \frac{1}{\sqrt{|h_1|^2 + |h_2|^2}} \begin{pmatrix} h_1^* \\ h_2^* \end{pmatrix}, \qquad y = h_1 v_1 x + h_2 v_2 x + w = \lVert\mathbf{h}\rVert\, x + w. \tag{16}$$

- Each path arrives with the real, positive gain $h_\ell v_\ell = |h_\ell|^2/\lVert\mathbf{h}\rVert$, so the two paths add in phase.
- **Equivalent channel:** $r = y$, gain $g = \lVert\mathbf{h}\rVert$, noise variance 1/SNR, output SNR $\lVert\mathbf{h}\rVert^2$SNR, the same as MRC with $L = 2$. Detection: $\hat{\imath} = \arg\min_i |y - \lVert\mathbf{h}\rVert u_i|^2$.
- **Scaling:** $\lVert\mathbf{v}\rVert = 1$, so the energy per channel use is $|v_1 x|^2 + |v_2 x|^2 = |x|^2$, with mean 1. No further scaling is needed.

4.1.3 Alamouti scheme

Transmission matrix. The transmitter does not know the channel. Two symbols are sent in two channel uses; rows are channel uses and columns are antennas:

$$\mathbf{S} = \frac{1}{\sqrt{2}} \begin{pmatrix} x_1 & x_2 \\ -x_2^* & x_1^* \end{pmatrix}, \qquad \begin{aligned} y_1 &= \tfrac{1}{\sqrt{2}}\,(h_1 x_1 + h_2 x_2) + w_1, \\ y_2 &= \tfrac{1}{\sqrt{2}}\,(-h_1 x_2^* + h_2 x_1^*) + w_2. \end{aligned} \tag{17}$$

The channel is constant over the two channel uses of a block and independent across blocks. Conjugating the second sample gives a linear model whose channel matrix has orthogonal columns:

$$\begin{pmatrix} y_1 \\ y_2^* \end{pmatrix} = \frac{1}{\sqrt{2}} \underbrace{\begin{pmatrix} h_1 & h_2 \\ h_2^* & -h_1^* \end{pmatrix}}_{\mathbf{H}} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} + \begin{pmatrix} w_1 \\ w_2^* \end{pmatrix}, \qquad \mathbf{H}^H\mathbf{H} = \lVert\mathbf{h}\rVert^2\, \mathbf{I}.$$

Combining. The receiver projects onto the two columns (normalised by $\lVert\mathbf{h}\rVert$):

$$\begin{aligned} r_1 &= \frac{h_1^* y_1 + h_2 y_2^*}{\lVert\mathbf{h}\rVert} = \frac{\lVert\mathbf{h}\rVert}{\sqrt{2}}\, x_1 + n_1, \\ r_2 &= \frac{h_2^* y_1 - h_1 y_2^*}{\lVert\mathbf{h}\rVert} = \frac{\lVert\mathbf{h}\rVert}{\sqrt{2}}\, x_2 + n_2. \end{aligned} \tag{18}$$

- The other symbol cancels exactly. The noises $n_1 = (h_1^* w_1 + h_2 w_2^*)/\|\mathbf{h}\|$ and $n_2 = (h_2^* w_1 - h_1 w_2^*)/\|\mathbf{h}\|$ are independent $\mathcal{CN}(0, 1/\text{SNR})$ (their covariance is $\mathbf{H}^H\mathbf{H}/(\|\mathbf{h}\|^2\text{SNR}) = \mathbf{I}/\text{SNR}$), so detecting x_1 and x_2 separately is ML.
- **Equivalent channel:** each r_k has gain $g = \|\mathbf{h}\|/\sqrt{2}$, noise variance $1/\text{SNR}$ and output SNR $\|\mathbf{h}\|^2\text{SNR}/2$, and is detected by the Problem 1 detector.
- **Scaling:** every entry of $\mathbf{S}$ carries $1/\sqrt{2}$, so a channel use carries $(|x_1|^2 + |x_2|^2)/2$, with mean 1, and each antenna carries half of it.
- The rate is one symbol per channel use, as for a single antenna.

4.1.4 Numerical check of the energy constraint

The simulation averages the energy actually sent, $\sum_\ell |s_\ell|^2$, over all channel uses at every SNR point (`results/q3/summary.txt`):

Scheme	Scaling	Total energy (min / max over SNR points)	Per antenna
Single antenna	$s = x$	1.000000 / 1.000000	1.0000
Beamforming	$s_\ell = (h_\ell^*/\|\mathbf{h}\|)\, x$	1.000000 / 1.000000	0.5000 / 0.5000
Alamouti	entries of $\mathbf{S}$, $\pm x_k^{(*)}/\sqrt{2}$	1.000000 / 1.000000	0.5000 / 0.5000

All schemes, and MRC (one transmit antenna sending x), send exactly the energy of the single-antenna system. For 4-QAM this holds in every single channel use, not just on average, because $|x|^2 = 1$ for every point.

4.2 Task 2: simulation

The script `problem3/transmit_diversity.py` runs Algorithm 1 for $\text{SNR}_{\text{dB}} \in \{0, 2, \ldots, 24\}$. For each scheme and SNR:

1. draw $N = 10^6$ equiprobable 4-QAM symbols and group them into blocks (one symbol per block, or two for Alamouti, giving $N/2 = 5 \times 10^5$ blocks);
2. draw one (h_1, h_2) per block and form the antenna signals with the scheme's transmitter: (16) for beamforming, (17) for Alamouti;
3. pass them through (15) and record the energy sent;
4. combine with the scheme's receiver, detect with the Problem 1 detector, and count the errors over all N symbols.

The MRC ($L = 2$) curve is simulated with the Problem 2 code. Fig. 6 shows all four curves on one figure, as the assignment asks; Fig. 7 compares the same points with the exact theory of Sec. 1.4. Every curve agrees with its theory (largest deviation 2.55σ), and beamforming agrees with the independent MRC simulation within 1.51σ.[1]

[1]The single-antenna curve is identical to the $L = 1$ curve of Problem 2. Both scripts start from seed 2026 and draw the symbols, channels and noise in the same order.

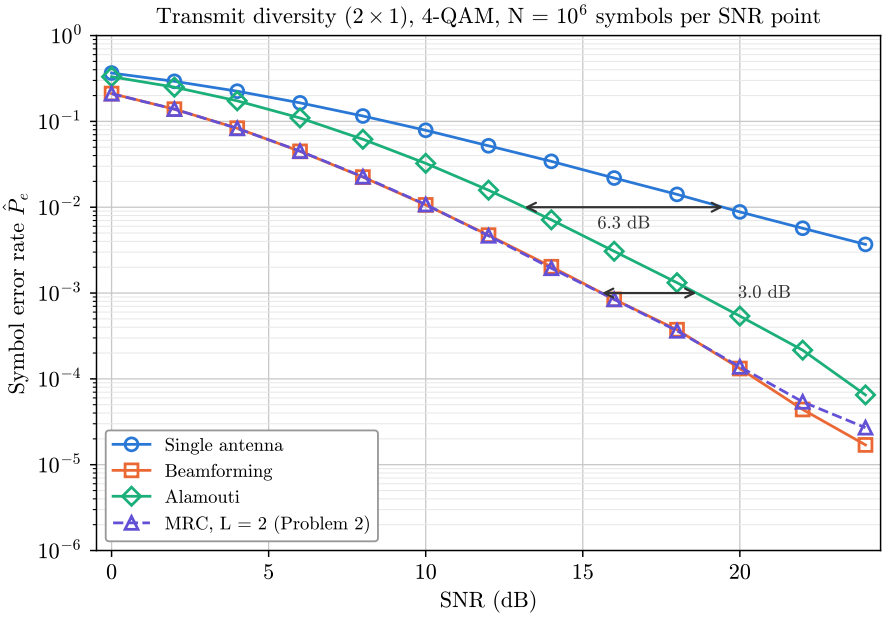


Figure 6: Problem 3: SER of 4-QAM with a single antenna, transmit beamforming and the Alamouti scheme (2×1), and MRC with $L = 2$ from Problem 2. All schemes transmit a total energy of 1 per channel use. Monte Carlo with $N = 10^6$ symbols per SNR point (Alamouti: 5×10^5 blocks of two symbols), $\mathrm{SNR}_{\mathrm{dB}} \in \{0, 2, \ldots, 24\}$; points with $n_{\mathrm{err}} < 10$ are omitted. Arrows: Alamouti is 3.0 dB behind beamforming at 10^{-3}, and the single antenna is 6.3 dB behind Alamouti at 10^{-2}. **Beamforming and MRC coincide; Alamouti has the same slope but is 3 dB behind; the gap to the single antenna grows as the SER falls** (Sec. 4.3.2).

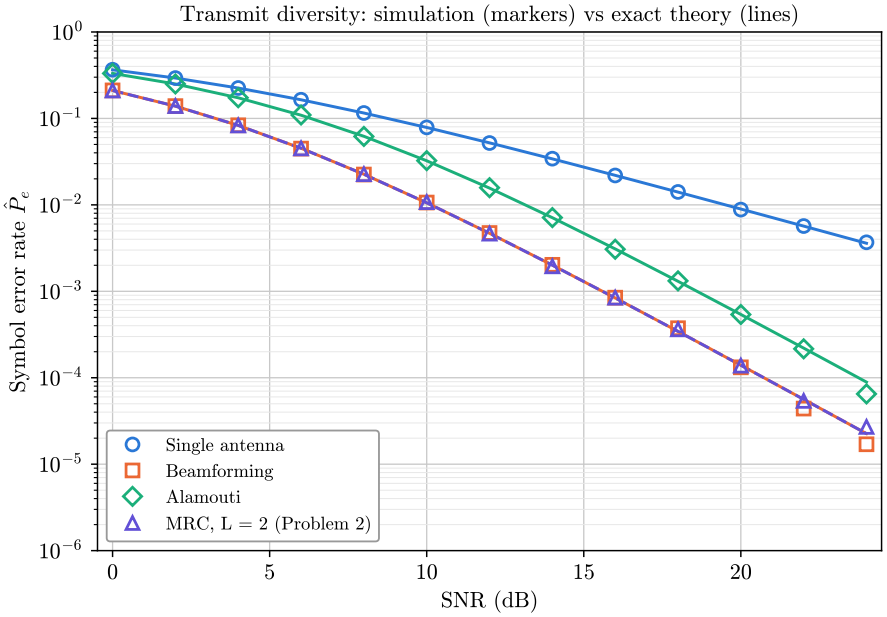


Figure 7: Extra figure: the simulated points of Fig. 6 ($N = 10^6$ symbols per SNR point, 0–24 dB) against the exact theory of Sec. 1.4 (lines). Beamforming and MRC have the same theoretical curve. **Every plotted point lies on its theory curve** (within 2.55σ).

4.3 Discussion

The assignment asks for two points: the diversity order of each scheme, read from the slope of its curve at high SNR (Sec. 4.3.1), and the SNR gaps between beamforming, Alamouti and MRC, with their origin (Sec. 4.3.2).

4.3.1 Diversity order

Table 5 gives the high-SNR slope of each curve.

Table 5: High-SNR slopes over the last four points with at least 50 errors, as in Table 3.

			Fitted slope		Exact theory,	Diversity
Scheme	Fit range	Local slopes	simulation	exact theory	60–70 dB	order
Single antenna	18–24 dB	1.02, 0.95, 0.94	0.97	0.99	1.00	1
Beamforming	14–20 dB	1.90, 1.77, 2.26	1.96	1.93	2.00	2
Alamouti	18–24 dB	1.94, 1.99, 2.61	2.16	1.94	2.00	2
MRC, $L = 2$	16–22 dB	1.83, 2.10, 2.04	2.00	1.96	2.00	2

- The single antenna has diversity order 1. Beamforming, Alamouti and MRC all have diversity order 2.
- In all three two-antenna schemes, the output SNR is proportional to $\|\mathbf{h}\|^2 = |h_1|^2 + |h_2|^2$. This is small only if *both* paths fade at once, which has probability $\approx \mathrm{SNR}^{-2}/2$.
- The Alamouti fit (2.16) is pulled up by its last step (2.61), from 216 to only 65 errors. One 2 dB step with so few errors is accurate to only about ± 0.3.
- Alamouti reaches full diversity even though the transmitter knows nothing about the channel.

4.3.2 SNR gaps and their origin

The arrows in Fig. 6 mark two of these gaps. Table 6 gives the SNR each scheme needs for a target SER.

Table 6: SNR (dB) needed for a target SER, read off the simulated curves (exact theory in brackets), and the gaps between the schemes. "–": the curve does not reach this SER within 0–24 dB.

	$P_\mathrm{e} = 10^{-2}$	$P_\mathrm{e} = 10^{-3}$	$P_\mathrm{e} = 10^{-4}$
Single antenna	19.46 (19.51)	– (29.58)	– (39.59)
Beamforming	10.15 (10.14)	15.61 (15.60)	20.51 (20.73)
Alamouti	13.15 (13.15)	18.62 (18.61)	23.28 (23.75)
MRC, $L = 2$	10.17 (10.14)	15.59 (15.60)	20.69 (20.73)
Beamforming − MRC	−0.02 (0.00)	+0.02 (0.00)	−0.18 (0.00)
Alamouti − MRC	+2.98 (3.01)	+3.03 (3.01)	+2.60 (3.01)
Alamouti − beamforming	+3.00 (3.01)	+3.01 (3.01)	+2.78 (3.01)
Single − Alamouti	+6.32 (6.36)	– (10.97)	– (15.84)

The 10^{-4} column is the least accurate (about ± 0.3 dB): there the curves are interpolated between points with only 44–216 errors.

- **Beamforming = MRC ($L = 2$): constant 0 dB.** Both give the output SNR $\|\mathbf{h}\|^2$SNR.
  - With MRC, the receiver collects the signal on two antennas, each with its own independent noise.

  - With beamforming, the transmitter uses its channel knowledge to make the two paths add coherently at the single receive antenna.
  - Both get the same array gain of 2 (3.01 dB, measured 3.01 dB) and diversity 2. The measured differences (≤ 0.18 dB) are Monte Carlo noise. The difference lies in what each needs: MRC needs two receive chains, beamforming needs channel feedback.
- **Alamouti: a constant 3 dB behind both** (exactly $10\log_{10} 2$).
  - Without channel knowledge, the transmitter cannot phase-align the two antennas. It must split its energy equally ($1/\sqrt{2}$ per antenna), and the two signals do not add coherently.
  - The output SNR is $\|\mathbf{h}\|^2\text{SNR}/2$, whose mean equals the SNR: array gain 1 (0 dB, measured 0.00 dB), but full diversity 2.
  - This SNR has exactly the beamforming/MRC distribution scaled by 1/2, so

$$P_{\text{e}}^{\text{Alamouti}}(\text{SNR}) = P_{\text{e}}^{\text{BF}}(\text{SNR}/2).$$

    The Alamouti curve is the beamforming curve shifted right by 3.01 dB at *every* SNR, not only asymptotically. The simulation gives 3.00 and 3.01 dB at 10^{-2} and 10^{-3}.
  - These 3 dB are the price of having no channel knowledge at the transmitter. In return, Alamouti needs no feedback, still sends one symbol per channel use, and is decoded with simple linear processing.
- **Single antenna versus the two-antenna schemes: not constant.** The slopes differ (1 against 2), so the gap grows as the SER falls. Relative to Alamouti it is 6.3 dB at 10^{-2}, and (from exact theory) 11.0 dB at 10^{-3} and 15.8 dB at 10^{-4}. This is a diversity gain, not an array gain: both schemes have array gain 1.

5 Summary of the gaps marked in the figures

Where a gap changes, the two values are those of the two arrows in the figure.

Curves compared	Fig.	Gap	Constant?	Reason
BPSK vs 4-QAM, AWGN	1	3.9, 3.3 dB	No	tends to 3 dB (half energy per component)
BPSK vs 4-QAM, fading	2	5.6, 5.5 dB	Yes	parallel slope-1 lines
AWGN vs fading, BPSK	3	17.3, 25.8 dB	No	exponential vs 1/SNR decay
MRC vs SC, $L = 1$	4	0 dB	Yes	identical receivers
MRC vs SC, $L = 2$	4	1.5 dB	Yes	array gain 3.01 vs 1.76 dB
MRC vs SC, $L = 4$	4	3.2 dB	Yes	array gain 6.02 vs 3.19 dB
Beamforming vs MRC, $L = 2$	6	0 dB	Yes	same output SNR
Alamouti vs beamforming	6	3.0 dB	Yes	no channel knowledge, so no array gain
Single antenna vs Alamouti	6	6.3 dB	No	diversity 1 vs 2: the gap grows

In short: curves with the same diversity order are parallel at high SNR and differ by a constant gap (array gain or modulation); curves with different diversity orders move apart as the SER falls.

A Numerical results

The tables are read directly from the CSV files in `results/` when the report is compiled. For Problems 2 and 3 they come from the `*_theory.csv` files, which hold the simulated values and the exact theory side by side. "sim." is n_{err}/N with $N = 10^6$. For points with fewer than 10 errors the raw error count is shown instead (in italics); these points are not plotted.

Table 7: Problem 1, AWGN: simulated SER and theory.

	BPSK		4-QAM	
$\mathrm{SNR}_{\mathrm{dB}}$	sim.	theory	sim.	theory
0	$7.93 \cdot 10^{-2}$	$7.86 \cdot 10^{-2}$	$2.93 \cdot 10^{-1}$	$2.92 \cdot 10^{-1}$
1	$5.66 \cdot 10^{-2}$	$5.63 \cdot 10^{-2}$	$2.45 \cdot 10^{-1}$	$2.45 \cdot 10^{-1}$
2	$3.74 \cdot 10^{-2}$	$3.75 \cdot 10^{-2}$	$1.97 \cdot 10^{-1}$	$1.97 \cdot 10^{-1}$
3	$2.33 \cdot 10^{-2}$	$2.29 \cdot 10^{-2}$	$1.52 \cdot 10^{-1}$	$1.52 \cdot 10^{-1}$
4	$1.26 \cdot 10^{-2}$	$1.25 \cdot 10^{-2}$	$1.10 \cdot 10^{-1}$	$1.10 \cdot 10^{-1}$
5	$6.05 \cdot 10^{-3}$	$5.95 \cdot 10^{-3}$	$7.40 \cdot 10^{-2}$	$7.39 \cdot 10^{-2}$
6	$2.43 \cdot 10^{-3}$	$2.39 \cdot 10^{-3}$	$4.54 \cdot 10^{-2}$	$4.55 \cdot 10^{-2}$
7	$7.53 \cdot 10^{-4}$	$7.73 \cdot 10^{-4}$	$2.51 \cdot 10^{-2}$	$2.50 \cdot 10^{-2}$
8	$2.03 \cdot 10^{-4}$	$1.91 \cdot 10^{-4}$	$1.18 \cdot 10^{-2}$	$1.20 \cdot 10^{-2}$
9	$4.00 \cdot 10^{-5}$	$3.36 \cdot 10^{-5}$	$4.80 \cdot 10^{-3}$	$4.82 \cdot 10^{-3}$
10	*6 err.*	$3.87 \cdot 10^{-6}$	$1.54 \cdot 10^{-3}$	$1.56 \cdot 10^{-3}$
11	*0 err.*	$2.61 \cdot 10^{-7}$	$3.68 \cdot 10^{-4}$	$3.88 \cdot 10^{-4}$
12	*0 err.*	$9.01 \cdot 10^{-9}$	$6.30 \cdot 10^{-5}$	$6.86 \cdot 10^{-5}$

Table 8: Problem 1, Rayleigh fading: simulated SER and theory. For 4-QAM both the table approximation and the exact value are listed.

	BPSK		4-QAM		
$\mathrm{SNR}_{\mathrm{dB}}$	sim.	theory	sim.	table approx.	exact
0	$1.47 \cdot 10^{-1}$	$1.46 \cdot 10^{-1}$	$3.65 \cdot 10^{-1}$	$4.23 \cdot 10^{-1}$	$3.65 \cdot 10^{-1}$
2	$1.08 \cdot 10^{-1}$	$1.08 \cdot 10^{-1}$	$2.93 \cdot 10^{-1}$	$3.35 \cdot 10^{-1}$	$2.93 \cdot 10^{-1}$
4	$7.72 \cdot 10^{-2}$	$7.71 \cdot 10^{-2}$	$2.25 \cdot 10^{-1}$	$2.54 \cdot 10^{-1}$	$2.25 \cdot 10^{-1}$
6	$5.37 \cdot 10^{-2}$	$5.30 \cdot 10^{-2}$	$1.64 \cdot 10^{-1}$	$1.84 \cdot 10^{-1}$	$1.64 \cdot 10^{-1}$
8	$3.54 \cdot 10^{-2}$	$3.55 \cdot 10^{-2}$	$1.15 \cdot 10^{-1}$	$1.29 \cdot 10^{-1}$	$1.15 \cdot 10^{-1}$
10	$2.32 \cdot 10^{-2}$	$2.33 \cdot 10^{-2}$	$7.87 \cdot 10^{-2}$	$8.71 \cdot 10^{-2}$	$7.86 \cdot 10^{-2}$
12	$1.49 \cdot 10^{-2}$	$1.51 \cdot 10^{-2}$	$5.20 \cdot 10^{-2}$	$5.77 \cdot 10^{-2}$	$5.22 \cdot 10^{-2}$
14	$9.89 \cdot 10^{-3}$	$9.67 \cdot 10^{-3}$	$3.40 \cdot 10^{-2}$	$3.76 \cdot 10^{-2}$	$3.41 \cdot 10^{-2}$
16	$6.16 \cdot 10^{-3}$	$6.16 \cdot 10^{-3}$	$2.21 \cdot 10^{-2}$	$2.42 \cdot 10^{-2}$	$2.20 \cdot 10^{-2}$
18	$3.96 \cdot 10^{-3}$	$3.92 \cdot 10^{-3}$	$1.42 \cdot 10^{-2}$	$1.55 \cdot 10^{-2}$	$1.41 \cdot 10^{-2}$
20	$2.50 \cdot 10^{-3}$	$2.48 \cdot 10^{-3}$	$8.92 \cdot 10^{-3}$	$9.85 \cdot 10^{-3}$	$8.95 \cdot 10^{-3}$
22	$1.58 \cdot 10^{-3}$	$1.57 \cdot 10^{-3}$	$5.81 \cdot 10^{-3}$	$6.25 \cdot 10^{-3}$	$5.68 \cdot 10^{-3}$
24	$1.01 \cdot 10^{-3}$	$9.92 \cdot 10^{-4}$	$3.60 \cdot 10^{-3}$	$3.96 \cdot 10^{-3}$	$3.60 \cdot 10^{-3}$
26	$6.35 \cdot 10^{-4}$	$6.27 \cdot 10^{-4}$	$2.29 \cdot 10^{-3}$	$2.50 \cdot 10^{-3}$	$2.27 \cdot 10^{-3}$
28	$4.21 \cdot 10^{-4}$	$3.96 \cdot 10^{-4}$	$1.41 \cdot 10^{-3}$	$1.58 \cdot 10^{-3}$	$1.44 \cdot 10^{-3}$
30	$2.84 \cdot 10^{-4}$	$2.50 \cdot 10^{-4}$	$9.13 \cdot 10^{-4}$	$9.99 \cdot 10^{-4}$	$9.08 \cdot 10^{-4}$
32	$1.48 \cdot 10^{-4}$	$1.58 \cdot 10^{-4}$	$5.33 \cdot 10^{-4}$	$6.30 \cdot 10^{-4}$	$5.73 \cdot 10^{-4}$
34	$1.06 \cdot 10^{-4}$	$9.95 \cdot 10^{-5}$	$3.42 \cdot 10^{-4}$	$3.98 \cdot 10^{-4}$	$3.62 \cdot 10^{-4}$
36	$5.70 \cdot 10^{-5}$	$6.28 \cdot 10^{-5}$	$2.25 \cdot 10^{-4}$	$2.51 \cdot 10^{-4}$	$2.28 \cdot 10^{-4}$
38	$4.40 \cdot 10^{-5}$	$3.96 \cdot 10^{-5}$	$1.38 \cdot 10^{-4}$	$1.58 \cdot 10^{-4}$	$1.44 \cdot 10^{-4}$
40	$2.50 \cdot 10^{-5}$	$2.50 \cdot 10^{-5}$	$8.40 \cdot 10^{-5}$	$1.00 \cdot 10^{-4}$	$9.09 \cdot 10^{-5}$

Table 9: Problem 2: simulated 4-QAM SER and exact theory. For $L = 1$, MRC and SC give identical results, equal to the single-antenna columns of Table 10.

	MRC, $L = 2$		SC, $L = 2$		MRC, $L = 4$		SC, $L = 4$	
$\mathrm{SNR}_{\mathrm{dB}}$	sim.	theory	sim.	theory	sim.	theory	sim.	theory
0	$2.10 \cdot 10^{-1}$	$2.10 \cdot 10^{-1}$	$2.64 \cdot 10^{-1}$	$2.64 \cdot 10^{-1}$	$7.70 \cdot 10^{-2}$	$7.73 \cdot 10^{-2}$	$1.82 \cdot 10^{-1}$	$1.82 \cdot 10^{-1}$
2	$1.39 \cdot 10^{-1}$	$1.39 \cdot 10^{-1}$	$1.86 \cdot 10^{-1}$	$1.86 \cdot 10^{-1}$	$3.48 \cdot 10^{-2}$	$3.49 \cdot 10^{-2}$	$1.09 \cdot 10^{-1}$	$1.09 \cdot 10^{-1}$
4	$8.29 \cdot 10^{-2}$	$8.30 \cdot 10^{-2}$	$1.20 \cdot 10^{-1}$	$1.20 \cdot 10^{-1}$	$1.27 \cdot 10^{-2}$	$1.29 \cdot 10^{-2}$	$5.58 \cdot 10^{-2}$	$5.57 \cdot 10^{-2}$
6	$4.54 \cdot 10^{-2}$	$4.52 \cdot 10^{-2}$	$7.02 \cdot 10^{-2}$	$7.01 \cdot 10^{-2}$	$3.83 \cdot 10^{-3}$	$3.94 \cdot 10^{-3}$	$2.39 \cdot 10^{-2}$	$2.38 \cdot 10^{-2}$
8	$2.26 \cdot 10^{-2}$	$2.26 \cdot 10^{-2}$	$3.76 \cdot 10^{-2}$	$3.75 \cdot 10^{-2}$	$1.07 \cdot 10^{-3}$	$1.01 \cdot 10^{-3}$	$8.36 \cdot 10^{-3}$	$8.41 \cdot 10^{-3}$
10	$1.05 \cdot 10^{-2}$	$1.06 \cdot 10^{-2}$	$1.85 \cdot 10^{-2}$	$1.85 \cdot 10^{-2}$	$2.23 \cdot 10^{-4}$	$2.24 \cdot 10^{-4}$	$2.54 \cdot 10^{-3}$	$2.48 \cdot 10^{-3}$
12	$4.76 \cdot 10^{-3}$	$4.69 \cdot 10^{-3}$	$8.69 \cdot 10^{-3}$	$8.56 \cdot 10^{-3}$	$3.90 \cdot 10^{-5}$	$4.48 \cdot 10^{-5}$	$5.95 \cdot 10^{-4}$	$6.23 \cdot 10^{-4}$
14	$2.05 \cdot 10^{-3}$	$2.01 \cdot 10^{-3}$	$3.77 \cdot 10^{-3}$	$3.78 \cdot 10^{-3}$	$1.00 \cdot 10^{-5}$	$8.27 \cdot 10^{-6}$	$1.35 \cdot 10^{-4}$	$1.37 \cdot 10^{-4}$
16	$8.33 \cdot 10^{-4}$	$8.38 \cdot 10^{-4}$	$1.62 \cdot 10^{-3}$	$1.61 \cdot 10^{-3}$	*2 err.*	$1.45 \cdot 10^{-6}$	$2.90 \cdot 10^{-5}$	$2.71 \cdot 10^{-5}$
18	$3.37 \cdot 10^{-4}$	$3.44 \cdot 10^{-4}$	$6.50 \cdot 10^{-4}$	$6.70 \cdot 10^{-4}$	*1 err.*	$2.45 \cdot 10^{-7}$	*5 err.*	$5.00 \cdot 10^{-6}$
20	$1.30 \cdot 10^{-4}$	$1.40 \cdot 10^{-4}$	$2.86 \cdot 10^{-4}$	$2.74 \cdot 10^{-4}$	*0 err.*	$4.04 \cdot 10^{-8}$	*0 err.*	$8.74 \cdot 10^{-7}$
22	$5.20 \cdot 10^{-5}$	$5.62 \cdot 10^{-5}$	$9.40 \cdot 10^{-5}$	$1.11 \cdot 10^{-4}$	*0 err.*	$6.58 \cdot 10^{-9}$	*0 err.*	$1.48 \cdot 10^{-7}$
24	$2.10 \cdot 10^{-5}$	$2.26 \cdot 10^{-5}$	$3.70 \cdot 10^{-5}$	$4.48 \cdot 10^{-5}$	*0 err.*	$1.06 \cdot 10^{-9}$	*0 err.*	$2.44 \cdot 10^{-8}$

Table 10: Problem 3: simulated 4-QAM SER and exact theory.

	Single antenna		Beamforming		Alamouti		MRC, $L = 2$	
$\mathrm{SNR}_{\mathrm{dB}}$	sim.	theory	sim.	theory	sim.	theory	sim.	theory
0	$3.66 \cdot 10^{-1}$	$3.65 \cdot 10^{-1}$	$2.11 \cdot 10^{-1}$	$2.10 \cdot 10^{-1}$	$3.32 \cdot 10^{-1}$	$3.32 \cdot 10^{-1}$	$2.10 \cdot 10^{-1}$	$2.10 \cdot 10^{-1}$
2	$2.93 \cdot 10^{-1}$	$2.93 \cdot 10^{-1}$	$1.39 \cdot 10^{-1}$	$1.39 \cdot 10^{-1}$	$2.51 \cdot 10^{-1}$	$2.50 \cdot 10^{-1}$	$1.39 \cdot 10^{-1}$	$1.39 \cdot 10^{-1}$
4	$2.25 \cdot 10^{-1}$	$2.25 \cdot 10^{-1}$	$8.33 \cdot 10^{-2}$	$8.30 \cdot 10^{-2}$	$1.74 \cdot 10^{-1}$	$1.73 \cdot 10^{-1}$	$8.29 \cdot 10^{-2}$	$8.30 \cdot 10^{-2}$
6	$1.64 \cdot 10^{-1}$	$1.64 \cdot 10^{-1}$	$4.47 \cdot 10^{-2}$	$4.52 \cdot 10^{-2}$	$1.10 \cdot 10^{-1}$	$1.09 \cdot 10^{-1}$	$4.50 \cdot 10^{-2}$	$4.52 \cdot 10^{-2}$
8	$1.16 \cdot 10^{-1}$	$1.15 \cdot 10^{-1}$	$2.24 \cdot 10^{-2}$	$2.26 \cdot 10^{-2}$	$6.19 \cdot 10^{-2}$	$6.21 \cdot 10^{-2}$	$2.26 \cdot 10^{-2}$	$2.26 \cdot 10^{-2}$
10	$7.88 \cdot 10^{-2}$	$7.86 \cdot 10^{-2}$	$1.06 \cdot 10^{-2}$	$1.06 \cdot 10^{-2}$	$3.25 \cdot 10^{-2}$	$3.24 \cdot 10^{-2}$	$1.08 \cdot 10^{-2}$	$1.06 \cdot 10^{-2}$
12	$5.19 \cdot 10^{-2}$	$5.22 \cdot 10^{-2}$	$4.73 \cdot 10^{-3}$	$4.69 \cdot 10^{-3}$	$1.58 \cdot 10^{-2}$	$1.56 \cdot 10^{-2}$	$4.65 \cdot 10^{-3}$	$4.69 \cdot 10^{-3}$
14	$3.42 \cdot 10^{-2}$	$3.41 \cdot 10^{-2}$	$2.03 \cdot 10^{-3}$	$2.01 \cdot 10^{-3}$	$7.13 \cdot 10^{-3}$	$7.11 \cdot 10^{-3}$	$1.94 \cdot 10^{-3}$	$2.01 \cdot 10^{-3}$
16	$2.19 \cdot 10^{-2}$	$2.20 \cdot 10^{-2}$	$8.44 \cdot 10^{-4}$	$8.38 \cdot 10^{-4}$	$3.07 \cdot 10^{-3}$	$3.09 \cdot 10^{-3}$	$8.44 \cdot 10^{-4}$	$8.38 \cdot 10^{-4}$
18	$1.41 \cdot 10^{-2}$	$1.41 \cdot 10^{-2}$	$3.74 \cdot 10^{-4}$	$3.44 \cdot 10^{-4}$	$1.32 \cdot 10^{-3}$	$1.31 \cdot 10^{-3}$	$3.63 \cdot 10^{-4}$	$3.44 \cdot 10^{-4}$
20	$8.82 \cdot 10^{-3}$	$8.95 \cdot 10^{-3}$	$1.32 \cdot 10^{-4}$	$1.40 \cdot 10^{-4}$	$5.39 \cdot 10^{-4}$	$5.40 \cdot 10^{-4}$	$1.38 \cdot 10^{-4}$	$1.40 \cdot 10^{-4}$
22	$5.69 \cdot 10^{-3}$	$5.68 \cdot 10^{-3}$	$4.40 \cdot 10^{-5}$	$5.62 \cdot 10^{-5}$	$2.16 \cdot 10^{-4}$	$2.20 \cdot 10^{-4}$	$5.40 \cdot 10^{-5}$	$5.62 \cdot 10^{-5}$
24	$3.69 \cdot 10^{-3}$	$3.60 \cdot 10^{-3}$	$1.70 \cdot 10^{-5}$	$2.26 \cdot 10^{-5}$	$6.50 \cdot 10^{-5}$	$8.91 \cdot 10^{-5}$	$2.70 \cdot 10^{-5}$	$2.26 \cdot 10^{-5}$